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SUPERSEDING
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MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10559285 SELECT LOGIC

This specification is approved for use by the US Army Armament Research and Development Command (ARRADCOM) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Circuit Card Assembly: 10559285 Select Logic which is the A3 component of the Electronic Unit, 11743131.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-STD-45662	Calibration System Requirements

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810	Environmental Test Methods

DRAWINGS

ARRADCOM

10559285	Circuit Card Assembly - Select Logic

Inspection Equipment

11750202	Automatic Card and Unit Test Set
11750431	Adapter, Select Logic
11821458	Shock Adapter, Plate Vib Horiz-Vert Plane Shock-Z
11821520	Automatic Test Assembly Central Processor Unit
11821647	Vibration/Shock Test Fixture

Packaging Data Sheet

P10559285	Packaging of Circuit Card Assembly - Select Logic: 10559285

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The select logic circuit card assembly, herein referred to as the assembly, shall be manufactured in accordance with Drawing 10559285 and drawings pertaining thereto and, when assembled, shall meet the requirements of this specification. (See 4.5.1)

3.1.1 Function. The assembly shall provide the following functions:

	Designation
a.	$\overline{\text{SELECT}}$ logic
b.	SELECT enable logic
c.	Computer lock logic
d.	Regulation
e.	BORESIGHT output
f.	COMPUTE output

3.1.2 General specifications. The following provisions of MIL-F-13926 apply: (See 4.5.1)

a. Order of precedence
b. Dimensions of tolerances
c. Inorganic protective surface finishes
d. Part identification and marking
e. Workmanship

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

a.	Temperature	$73^{\circ} \pm 18^{\circ}F$
b.	Relative humidity	50 percent $\pm$ 30 percent
c.	Atmospheric pressure	28.5 + 2.0 - 3.0 in. Hg.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.2).

3.3 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under standard ambient conditions of 3.1.3.

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TABLE I. Loads, power and signals.

Item	Condition	Characteristic	Connections
1.0	Loads	Resistance (ohms, ± 5%).	Connected between following pins of P1:
1.1	Resistor	3000	22B and 15A
1.2	Resistor	3000	21A and 15A
1.3	Resistor	3000	20B and 15A
1.4	Resistor	1500	18B and 15A
1.5	Resistor	910	26B and 15A
1.6	Resistor	1800	28B and 36B
1.7	Resistor	2700	30B and 36B
1.8	Resistor	1300	As specified herein
1.9	Resistor	130	As specified herein
1.10	Resistor	51	As specified herein
1.11	Resistor	43	As specified herein
2.0	Power sources	Tolerance (volts) / Maximum ripple (mV p-p)	Applied between following pins of P1:
2.1	+5V	± 0.2 / 25	15A (+) and 16B (-)
2.2	-6V	± 0.2 / 25	33A (+) and 16B (-)
2.3	+15V	± 0.7 / 25	14B (+) and 36B (-)
2.4	+29V	± 1.0 / 25	32B (+) and 36B (-)
3.0	Signal sources		Applied to input terminals as specified herein
3.1	Digital type A	Logical one: 4 ± 1V Logical zero: .2 ± .2V	
3.2	Digital type B	Logical one: 15 ± 1V Logical zero: Open circuit	

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3.3.1 SELECT logic. With the logical zero type A signal of table 1 momentarily applied to P1 - 12B once as an initial condition, followed by the application of the digital type A signal of table 1 to each group of input pins for each item as specified in table II, the digital signal shall appear at each listed output pin as specified for that item. (See 4.6.2.1)

TABLE II. Select logic.

Item	Input pins of P1 1/							Output pins of P1 1/			
	1A	2B	3A	4B	6B	8B	5A	22B	21A	20B	10B
1	0	0	0	0	1	0	0	L	H	H	L
2	0	0	0	0	0	1	0	H	L	H	NA
3	0	0	0	0	0	0	1	L	H	H	NA
4	1	0	0	0	0	0	1	L	H	H	H
5	0	1	0	0	0	0	1	H	L	H	NA
6	1	1	0	0	0	0	1	H	H	L	NA
7	0	0	1	0	0	0	1	H	H	L	NA
8	0	0	0	1	0	0	1	H	H	L	NA

1/ Legend: 1 = Logical one, type A H = 4.2 ± 1.0V
0 = Logical zero, type A L = .2 ± .2V
NA = Not Applicable

3.3.2 SELECT enable logic. With the digital type A signal of table I applied to each group of input pins for each item as specified in table III, and each item performed in sequence, a digital signal shall appear at each listed output pin as specified for that item. (See 4.6.2.2)

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TABLE III. Select enable logic.

Item	Input Pins of P1 1/									Output Pins of P1 1/		
	1A	2B	3A	4B	6B	8B	5A	12B	7A	22B	21A	20B
1	1	1	0	0	1	0	0	0	1	H	NA	NA
2	1	1	0	0	1	0	0	1	1	L	NA	NA
3	1	1	0	0	0	1	0	0	1	NA	H	NA
4	1	1	0	0	0	1	0	1	1	NA	L	NA
5	1	1	0	0	0	0	1	0	1	NA	NA	H
6	1	1	0	0	0	0	1	1	1	NA	NA	L
7	1	1	0	0	0	0	1	1	0	NA	NA	H

1/ Legend: See table II.

3.3.3 Computer lock logic.

3.3.3.1 Multiple pin output. With the digital type A signal of table I applied to each group of input pins for each item as specified in table IV, followed by a logical zero type A signal momentarily applied to P1-7A and P1-12B in sequence for each item (P1-12B, item 1, excepted), digital signals shall appear at the output pins as specified in table IV. (See 4.6.2.3)

TABLE IV. Multiple pin output.

Item	Input Pins of P1 1/							Output Pins of P1 1/		
	1A	2B	3A	4B	11A	12B	23A	18B	24B	25A
1	1	1	1	1	1	0	0	H	H	H
2	1	1	1	1	1	1	0	L	L	H
3	0	0	0	1	1	1	0	NA	H	L
4	1	1	1	1	1	1	1	NA	H	NA
5	1	1	1	1	0	1	0	NA	H	NA

1/ Legend: See table II.

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3.3.3.2 Single pin output. With the digital type A signal of table I applied to each group of input pins for each item as specified in table V, followed by a logical zero type A signal momentarily applied to P1-7A and P1-12B in sequence for each item, a digital signal shall appear at the output pin as specified in table V.

TABLE V. Single pin output.

Item	Input Pins of P1 1/										Output 1/	
	1A	2B	3A	4B	5A	6B	8B	9A	11A	23A	31A	P1-24B
1	1	0	0	0	0	1	0	1	0	0	C	L
2	1	0	0	0	0	1	0	1	0	1	C	H
3	1	0	0	0	0	1	0	0	0	0	C	H
4	1	0	0	0	0	1	0	1	0	0	D	H
5	0	1	0	0	1	0	0	1	0	0	C	L
6	1	1	0	0	1	0	0	1	0	0	C	L
7	0	0	1	0	1	0	0	1	0	0	C	H
8	1	1	0	0	0	1	0	1	0	0	C	H
9	1	1	0	0	0	0	1	1	0	0	C	H

1/ Legend: 1 = logical one, type A C = logical one, type B H = 4.2 ± 1.0V
0 = logical zero, type A D = logical zero, type B L = .2 ± .2V

3.3.4 Regulation.

3.3.4.1 High resistance. With the 1300-ohm load specified in table I connected between P1-19A and P1-14B, the voltage at P1-19A shall be 7.5 ± 0.7 Vdc. (See 4.6.2.4)

3.3.4.2 Low resistance. With the 130-ohm load specified in table I connected between P1-19A and P1-14B, the voltage at P1-19A shall be 7.5 ± 0.7 Vdc.

3.3.5 BORESIGHT output.

3.3.5.1 High voltage. With the type A logical zero and type A logical one signals of table I applied to P1-7A and P1-23A, respectively, the voltages at P1-28B and P1-26B shall be 28.5 ±1.5 Vdc and 4.2 ±1.0 Vdc, respectively.

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3.3.5.2 Low voltage. With the conditions of 3.3.5.1 and with the 51-ohm load of table I connected in parallel with the 1800-ohm load between P1-28B and P1-36B, the voltage at P1-26B shall be 0.2 ±0.2 Vdc. (See 4.6.2.5)

3.3.6 COMPUTE output.

3.3.6.1 Double voltage. With the type A logical zero signal of table 1 applied to P1-18B, the voltages at P1-30B and P1-26B shall be 28.5 ±1.5 Vdc and 4.2 ±1.0 Vdc, respectively.

3.3.6.2 Single voltage. With the conditions of 3.3.6.1 and with the 43-ohm load of table I connected in parallel with the 620-ohm load between P1-30B and P1-36B, the voltage at P1-26B shall be 0.2 ±0.2 Vdc. (See 4.6.2.6)

3.4 Environmental.

3.4.1 Shock. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to three half sine wave shock pulses of 40 ± 4 gravity units (g) for a duration of 18 ± 3 milliseconds (msec) applied in each direction along three mutually perpendicular axes. In addition, the assembly shall be capable of withstanding three half sine wave shock pulses of 100.0 ± 10.0 g for a duration of 1.5 ± 0.2 msec applied in each direction along three mutually perpendicular axes. (See 4.6.3.1)

3.4.2 Vibration. The assembly shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to the vibration profile of figure 1. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes. (See 4.6.3.2)

3.4.3 Operating temperature. The assembly shall be capable of operating as specified herein over the operating temperature range of -25°F to +140°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5)

3.4.4 Storage temperature. The assembly shall be capable of operating as specified herein after exposure to storage temperatures ranging from -65°F to +160°F. (See 4.6.3.3, 4.6.3.4 and 4.6.3.5)

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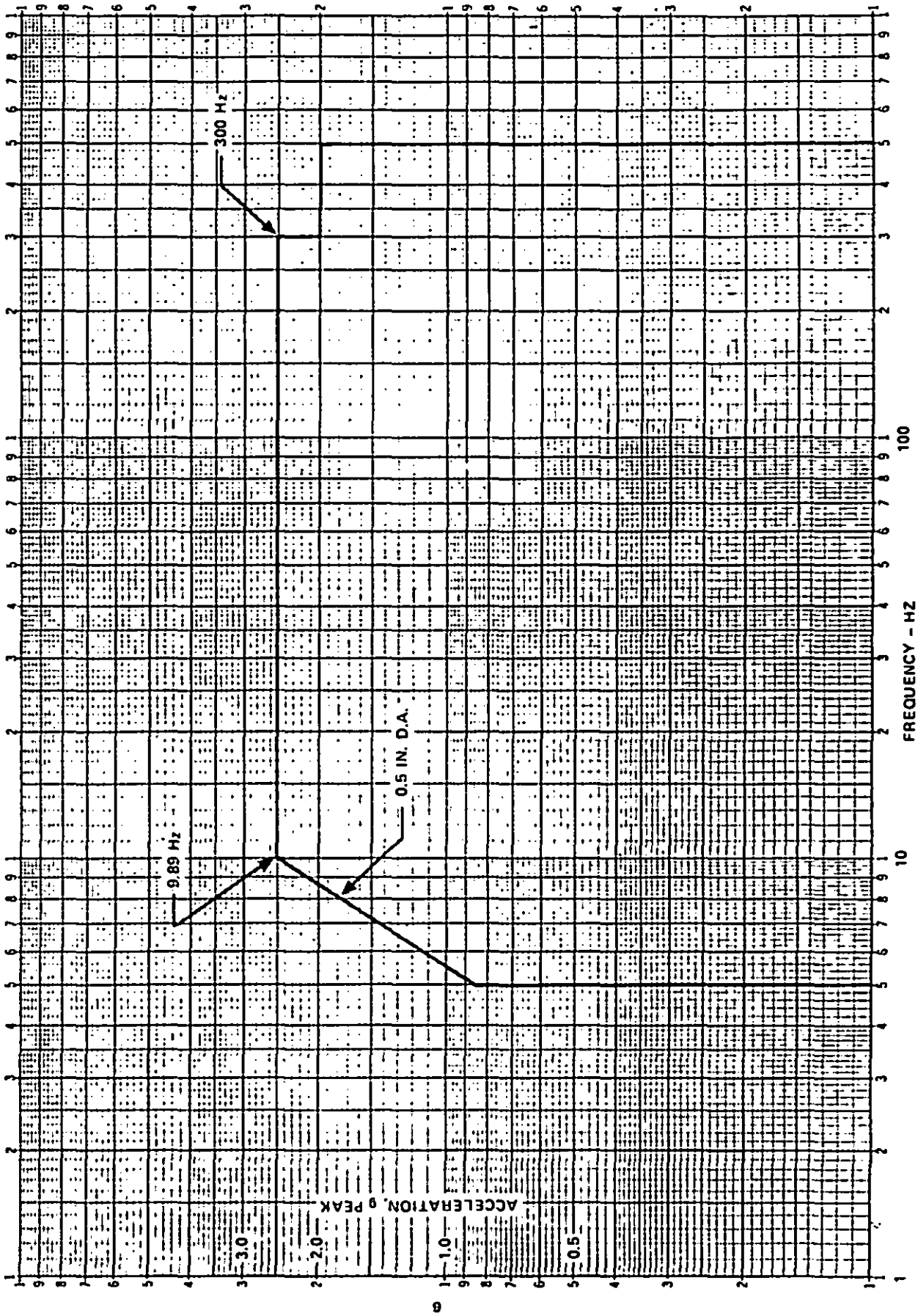


FIGURE 1. Vibration profile.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4).
2. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

4.4 First article. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three assemblies selected at random by the Government representative from the first fifteen production assemblies.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in tables VI, VII and VIII.

4.4.3 Failure. Failure of any assembly to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any assembly to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.

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TABLE VI. Performance tests.

Item	Characteristic	Requirement	Test Procedure
101	$\overline{\text{SELECT}}$ logic	3.3.1	4.6.2.1
102	SELECT enable logic	3.3.2	4.6.2.2
103	Computer lock logic	3.3.3	4.6.2.3
104	Regulation	3.3.4	4.6.2.4
105	BORESIGHT output	3.3.5	4.6.2.5
106	COMPUTE output	3.3.6	4.6.2.6

TABLE VII. General tests.

Item	Characteristic	Requirement	Test Procedure
301	Fabrication	3.1	Applicable drawings - Visual
302	General Specification	3.1.2	MIL-F-13926 - Visual

TABLE VIII. Environmental tests.

Item	Characteristic	Requirement	Test Procedure
303	Shock	3.4.1	4.6.3.1
304	Vibration	3.4.2	4.6.3.2
305	High Temperature	3.4.3 and 3.4.4	4.6.3.3
306	Low Temperature	3.4.3 and 3.4.4	4.6.3.4
307	Temperature cycling	3.4.3 and 3.4.4	4.6.3.5

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4.5 Quality conformance inspection.

4.5.1 Procurement conditions. The following inspection plans shall apply where the assembly is procured for use as a:

Use	Plan
a. Logistics spare assembly apart from the next higher assembly	A and C
b. Component assembly of the next higher assembly	B and C

4.5.2 Inspection plan A.

4.5.2.1 General sample and tests. One assembly, as a control sample, shall be selected at random by the Government representative from each 100 assemblies produced and shall be subjected to all the tests in table VII.

4.5.2.2 Environmental sample and tests. Three assemblies, as a control sample, shall be selected at random by the Government representative from each 50 assemblies produced or from each month's production, whichever occurs first. All the tests in table VIII shall be applied separately to each assembly in the sample.

4.6.2.3 Acceptance. Where any one assembly of either sample fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subject to the provisions of MIL-STD-105.

4.5.3 Inspection plan B.

4.5.3.1 Sample and tests. The sample shall be selected by the Government representative in accordance with the provisions of MIL-STD-105. All the tests in table VII and only the test of item 307 in table VIII shall be applied.

4.5.3.2 Acceptance. Acceptance and rejection shall be in accordance with MIL-STD-105.

4.5.4 Inspection plan C.

4.5.4.1 Sample and tests. Each assembly in every lot shall be subjected to all the tests in table VI and shall be examined visually for completeness, improper assembly and evidence of poor workmanship.

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4.5.4.2 Acceptance. Where any one assembly fails to meet any specified requirement, the defective assembly shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.5 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and MIL-STD-45662.

4.5.6 Packaging inspection. The sampling and inspection of the preservation - packaging, packing and container marking shall be in accordance with the provisions of packaging data sheet P10559285.

4.6 Methods of inspection.

4.6.1 Test equipment. Unless the assembly is procured apart from the next higher assembly, the test equipment in table IX shall be used where specified to perform the required test.

TABLE IX. Test equipment.

Item	Part Number	Nomenclature
1	11750202	Automatic Card and Unit Test Set
2	11750431	Adapter, Select Logic
3	11821458	Shock Adapter, Plate Vib Horiz - Vert Plane Shock - Z
4	11821520	Automatic Test Assembly Central Processor Unit
5	11821647	Vibration/Shock Test Fixture

4.6.2 Performance tests.

4.6.2.1 SELECT logic. Use items 1, 2 and 4 of table IX or suitable equipment to determine compliance with 3.3.1.

4.6.2.2 SELECT enable logic. Use items 1, 2 and 4 of table IX or suitable equipment to determine compliance with 3.3.2.

4.6.2.3 Computer lock logic. Use items 1, 2 and 4 of table IX or sutiable equipment to determine compliance with 3.3.3.

4.6.2.4 Regulation. Use items 1, 2 and 4 of table IX or suitable equipment to determine compliance with 3.3.4.

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4.6.2.5 BORESIGHT output. Use items 1, 2 and 4 of table IX or suitable equipment to determine compliance with 3.3.5.

4.6.2.6 COMPUTE output. Use items 1, 2 and 4 of table IX or suitable equipment to determine compliance with 3.3.6.

4.6.3 Environmental tests.

4.6.3.1 Shock.

4.6.3.1.1 Basic design. Use item 5 of table IX or mount the assembly in a suitable shock fixture and subject it to the shock test specified in MIL-STD-810, method 516.2, procedure I, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 40 ±4 g with a time duration of 18±3 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table VI.

4.6.3.1.2 High intensity. Use items 3 and 5 of table IX or mount the assembly in a suitable shock fixture and subject it to the shock test specified in MIL-STD-810, method 516.2, procedure IV, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude shall be 100.0 ±10.0 g with a time duration of 1.5 ±0.2 msec measured at the 10 percent amplitude points. At the conclusion of this test, subject the assembly to the performance tests of table VI.

4.6.3.2 Vibration. Vibration A as specified in 4.6.3.2.1 shall be conducted only for first article inspection. Vibration B as specified in 4.6.3.2.2 shall be conducted only for quality conformance inspection.

4.6.3.2.1 Vibration A. Use items 3 and 5 of table IX or mount the assembly in a suitable vibration fixture and subject the assembly to the vibration test specified in MIL-STD-810, method 514.2 procedure VIII, except that the test level shall be the vibration curve shown in figure 1 and duration of exposure shall be not less than 80 minutes per axis. At the conclusion of this test, subject the assembly to the performance tests of table VI.

4.6.3.2.2 Vibration B. This test shall be conducted as in 4.6.3.2.1 except that the period of vibration shall be not less than 15 minutes in each axis.

4.6.3.3 High temperature. Use item 5 of table IX or suitable equipment and subject the assembly to the high temperature test specified in MIL-STD-810, method 501.1, procedure I. The highest operating temperature shall be 140°F. The performance tests of table VI shall be applied.

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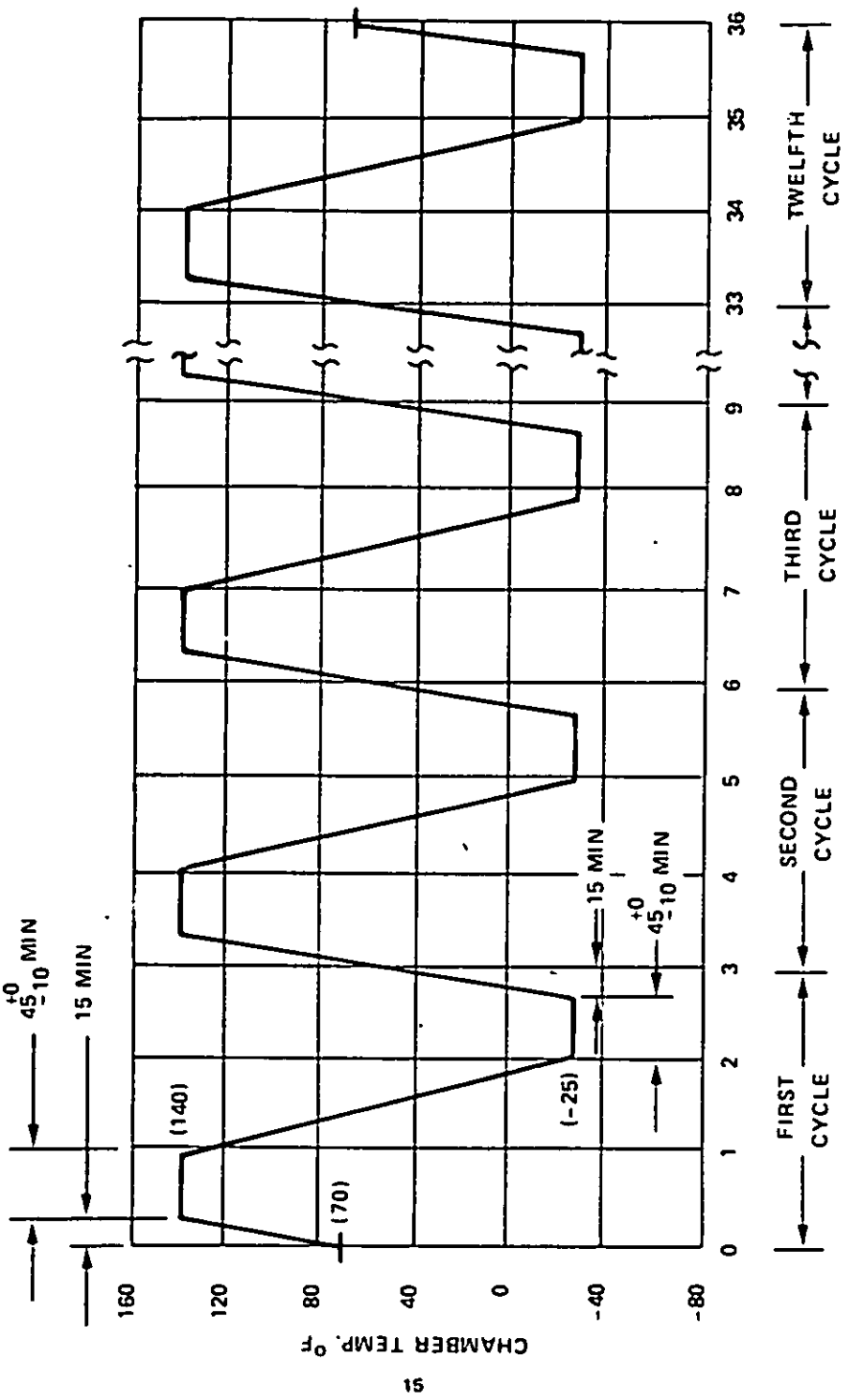


FIGURE 2. Temperature cycling.

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4.6.3.4 Low temperature. Use item 5 of table IX or suitable equipment and subject the assembly to the low temperature test specified in MIL-STD-810, method 502.1, procedure I. The storage temperature shall be -65°F. The lowest operating temperature shall be -25°F. The performance tests of table VI shall be applied.

4.6.3.5 Temperature cycling. With power and loads of table I applied, subject the assembly to the temperature profile specified in figure 2. At the conclusion of this test, subject the assembly to the performance tests of table VI.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with packaging data sheet P10559285. The level of protection shall be as specified in the procurement document. (See 4.5.6)

6. NOTES

6.1 Intended use. The select logic circuit card assembly is a functional module of the Laser Receiver-Transmitter, intended for use in the Laser Range Finder AN/VVS-1. The assembly contains the circuitry for counter selection, high-level line drivers, and low-level line drive clamping. (Common to AN/VVG-2 LRF)

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification

b. Applicable packaging data sheet number (see 5.1)

c. Selection of applicable levels of preservation, packaging and packing

d. Applicable stock number

e. Requirement for first article submission.

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army - AR

Preparing Activity:
Army - AR
Project No. 1240-A793

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